

3.1 Newton's Laws of Motion

Question Paper

Course	CIE A Level Physics
Section	3. Dynamics
Topic	3.1 Newton's Laws of Motion
Difficulty	Easy

Time allowed: 50

Score: /41

Percentage: /100

Question 1a

A brick of mass 2 kg is resting on the floor as shown in Fig. 1.1.

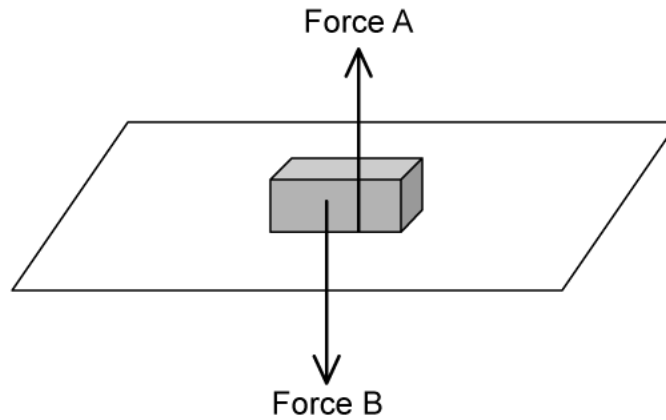


Fig. 1.1

(i)
Define equilibrium.

[1]

(ii)
State the names of forces A and B.

[1]

[2 marks]

Question 1b

(i)
State Newton's First Law of motion.

[1]

(ii)
Explain how Newton's First Law applies to the brick in Fig. 1.1.

[2]

[3 marks]

Question 1c

(i)

State Newton's Third Law of motion.

[1]

(ii)

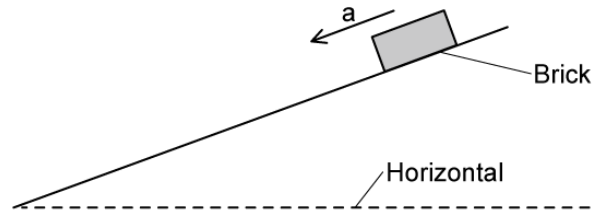
Explain how Newton's Third Law can be applied to **one** of the forces acting on the brick.

[3]

[4 marks]

Question 1d

The surface is raised at one end to form a slope as shown in Fig. 1.2. This causes the brick to accelerate at a constant rate of 0.25 m s^{-2} .

**Fig. 1.2**

(i)

State Newton's Second Law of motion.

[1]

(ii)

Use Newton's Second Law to determine the magnitude of the resultant force on the brick.

[2]

[3 marks]**Question 2a**

A skydiver of mass 75 kg jumps from an aircraft.

(i)

State the difference between the mass of a body and its weight.

[2]

(ii)

Calculate the weight of the skydiver.

[2]

[4 marks]

Question 2b

The graph in Fig. 1.1 shows the vertical velocity of the skydiver during the first 40 s of the fall before opening his parachute.

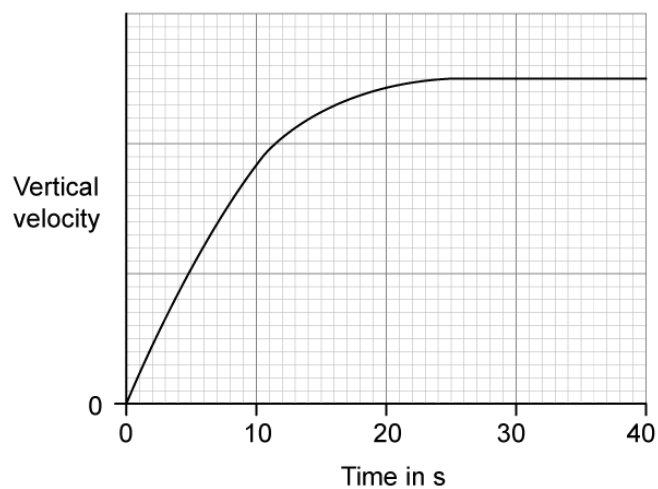


Fig. 1.1

Explain the shape of the graph

- (i)
between 0 and 5 s
- (ii)
between 5 and 25 s
- (iii)
between 25 and 40 s

[2]

[2]

[2]

[6 marks]

Question 2c

Fig. 1.2 shows the skydiver falling at a constant velocity.

Add labelled arrows to Fig. 1.2 to show the forces acting on the skydiver.



Fig. 1.2

[3 marks]

Question 2d

The skydiver opens his parachute after 40 s.

On Fig. 1.3 below, complete the graph to show how the skydiver's vertical velocity changes after 40 s.

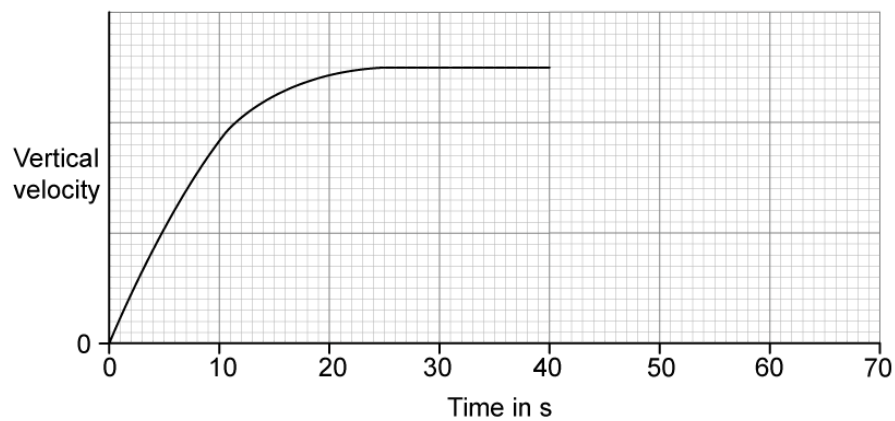


Fig. 1.3

[2 marks]

Question 3a

A hot-air balloon is tied to the ground by two ropes, as shown in Fig. 1.1.

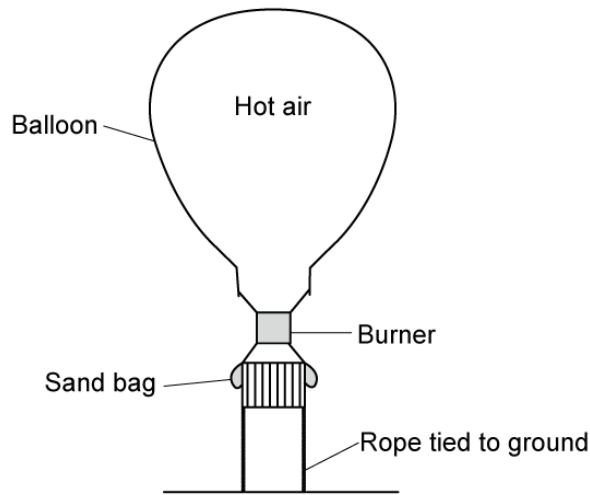


Fig. 1.1

Label Fig. 1.1 to show the forces acting on the balloon.

[3 marks]

Question 3b

The tension T required to keep the balloon fixed to the ground is 400 N in each rope.

The ropes are untied and the balloon starts to move upwards. The balloon and its contents have a total mass of 1500 kg.

As the balloon begins its ascent, calculate

(i)
the weight of the balloon,

[2]

(ii)
the resultant force acting on the balloon.

[2]

[4 marks]**Question 3c**

(i)

Calculate the initial acceleration of the balloon.

[2]

(ii)

Explain how the upward acceleration of the balloon changes during the first few seconds of its flight.

[3]**[5 marks]****Question 3d**

If the passengers want to change the upward acceleration, they can release some sand from the bags.

State and explain how this affects the upward acceleration of the balloon.

[2 marks]

Model Answers: Easy

1a

(a)

(i) Equilibrium is defined as:

- (When a body or system has) constant velocity / zero acceleration / no resultant force / no resultant torque; [1 mark]

(ii) Forces A and B are:

- A is the normal / reaction / contact force;
AND
- B is the weight (of the brick); [1 mark]

[Total: 2 marks]

1b

(b)

(i) Newton's First Law states:

- A body continues at rest or constant velocity unless acted on by a resultant (external) force; [1 mark]

(ii) Newton's First Law applies to the brick because:

Any **two** from:

- The brick is at rest / stationary; [1 mark]
- (Because) there is no resultant force acting on it; [1 mark]
- The upwards force = the downwards force **OR** force A = force B **OR** weight =

normal contact / reaction force; [1 mark]

[Total: 3 marks]

1c

(c)

(i) State Newton's Third Law:

- If body A exerts a force on body B, then body B will exert a force on body A of equal magnitude but in the opposite direction; [1 mark]

(ii) Explain how Newton's Third Law applies to one of the forces acting on the brick:

EITHER

- The downward force of gravity is exerted by the Earth on the brick; [1 mark]
- is equal and opposite to; [1 mark]
- the upwards force of gravity is exerted by the brick on the Earth; [1 mark]

OR

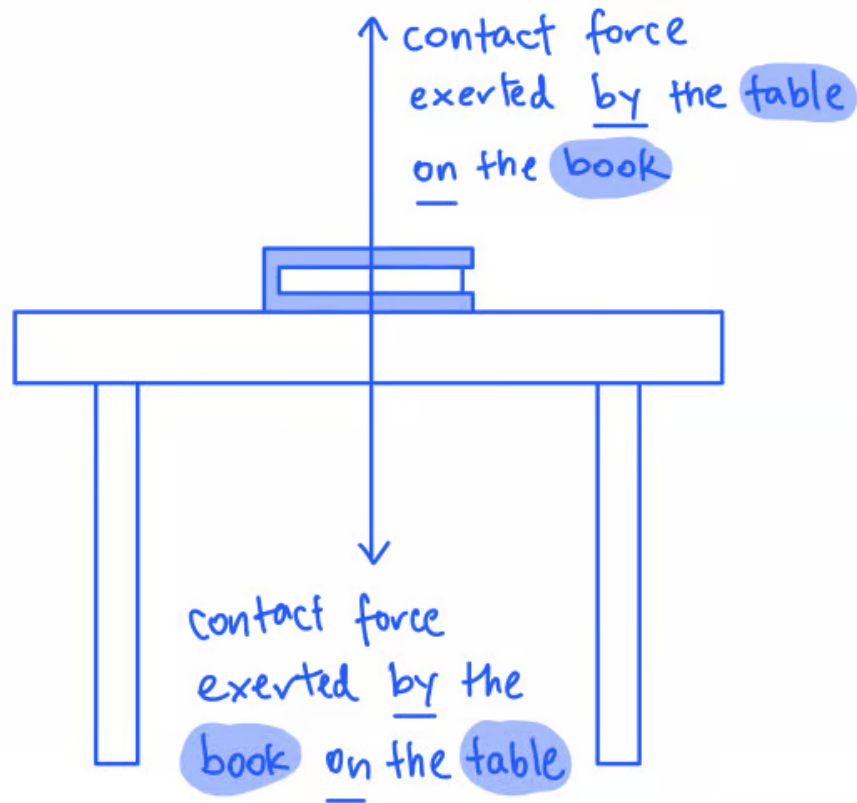
- The upwards contact force is exerted by the table on the brick; [1 mark]
- is equal and opposite to; [1 mark]
- the downward contact force is exerted by the brick on the table; [1 mark]

[Total: 4 marks]

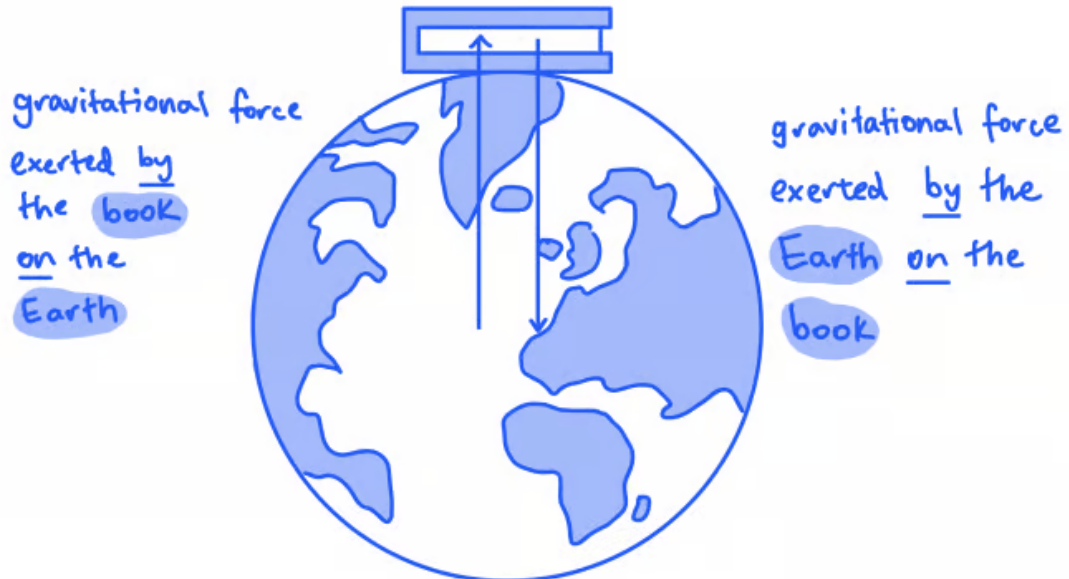
It is a common mistake to think that the reaction force and weight are a Newton's Third Law pair, make sure you understand why this is incorrect:

1. They both are acting on the same object (the brick)
2. They are not of the same type of force

The correct application of Newton's Third Law for the **contact force** is shown below:



The correct application of Newton's Third Law for **weight** is shown below:



1d

(d)

(i) Newton's Second Law states:

- The (resultant) force (on a body) is equal to its rate of change of momentum and in the same direction;
- OR**
- The (resultant) force (on a body) is directly proportional to its acceleration and in the same direction; [1 mark]

(ii) Determine the magnitude of the resultant force on the brick:

List the known quantities:

- Mass of the brick, $m = 2 \text{ kg}$
- Acceleration of the brick, $a = 0.25 \text{ m s}^{-2}$

Use Newton's second law to calculate the resultant force on the brick:

- Resultant force: $F = ma$
- $F = 2 \times 0.25$ [1 mark]
- $F = 0.5 \text{ N}$ [1 mark]

[Total: 3 marks]

2a

(a)

(i) The difference between the mass of a body and its weight is:

- Mass is the measure of a body's resistance, or inertia, to changes in velocity / motion; [1 mark]
- Weight is the effect of a gravitational field on mass **OR** weight is the force on a mass due to gravity; [1 mark]

Accept comparisons e.g. mass is constant, weight varies / mass is a scalar, weight is a vector

(ii) Calculate the weight of the skydiver:

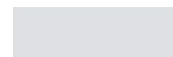
List the known quantities:

- Mass of the skydiver, $m = 75 \text{ kg}$
- Acceleration due to gravity, $g = 9.81 \text{ N kg}^{-1}$

Substitute the values into the equations:

- Weight: $W = mg$ [1 mark]
- $W = 75 \times 9.81$
- $W = 735.75 = 740 \text{ N}$ [1 mark]

[Total: 4 marks]



List the known quantities:

- Mass of the skydiver, $m = 75 \text{ kg}$
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Substitute the values into the equations:

- Weight: $W = mg$ [1 mark]
- $W = 75 \times 9.81$
- $W = 735.75 = 740 \text{ N}$ [1 mark]

[Total: 4 marks]

2b

(b) Explain the shape of the velocity-time graph:

(i) Between 0 and 5 s:

- Only (the skydiver's) weight acts / resultant force downwards; [1 mark]
- So, at first (the skydiver) accelerates; [1 mark]

(ii) Between 5 and 25 s:

Any **two** from:

- Drag / air resistance increases with speed; [1 mark]
- (Hence) as speed increases, the resultant force decreases; [1 mark]
- As resultant force decreases, acceleration decreases; [1 mark]

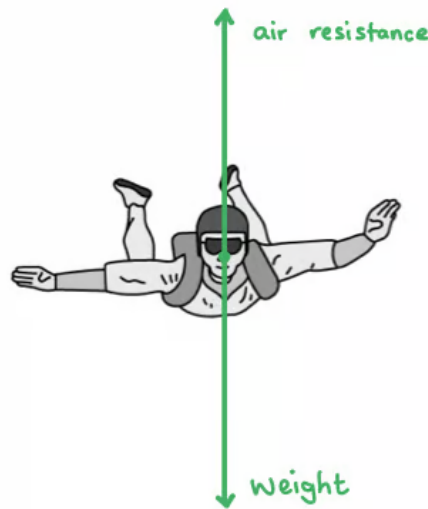
(iii) Between 25 and 40 s:

- The forces (of drag and weight) are equal / balanced; [1 mark]
- (This is the point at which) the skydiver reaches terminal velocity; [1 mark]

[Total: 6 marks]

2c

(c) The forces on the skydiver when travelling at constant velocity are:



- Arrow up labelled drag / air resistance / air friction; [1 mark]
- Arrow down labelled weight; [1 mark]
- Arrows drawn approximately equal length, or 740 N seen (value from **(a)**); [1 mark]

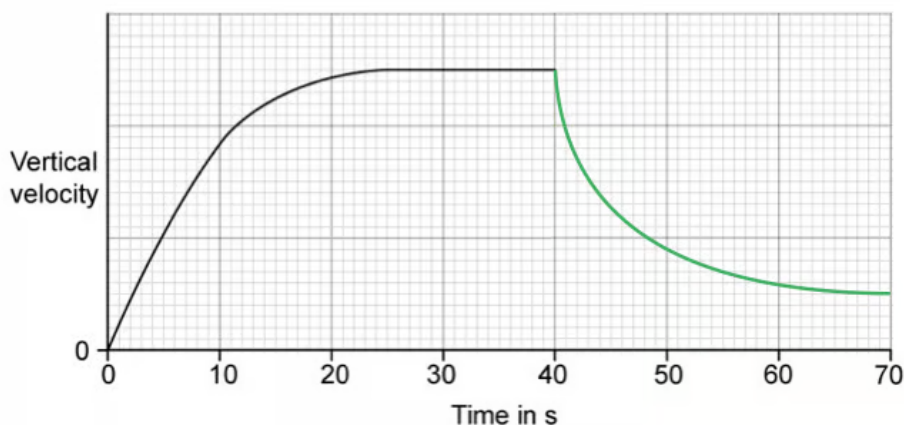
[Total: 3 marks]

In this question, you would score all the marks if your arrows were not touching.

However, it's more physically correct to draw the arrows from the same point, to indicate they are acting from the centre of gravity, so it's a good habit to get into!

2d

(d) The variation of vertical velocity after opening the parachute is as follows:

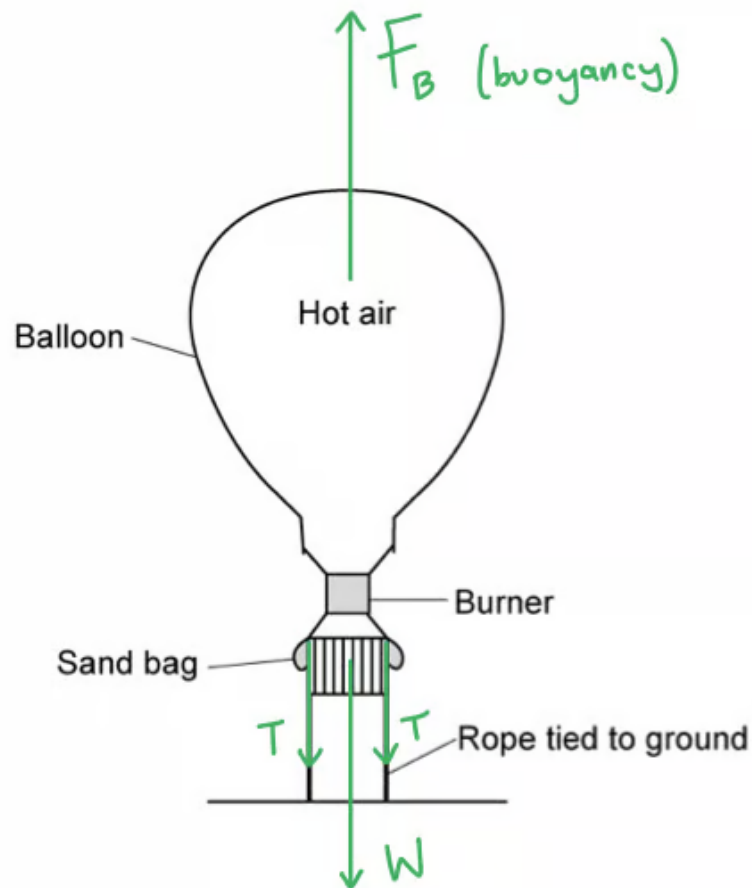


- Smooth curve drawn showing a decreasing gradient; [1 mark]
- Horizontal line drawn near end to show a new lower terminal velocity; [1 mark]

[Total: 2 marks]

3a

(a) The forces acting on the balloon are:



- Upward force / buoyancy force drawn; [1 mark]
- Weight / W acting downwards from centre; [1 mark]
- Two tension / T arrows acting downwards on ropes; [1 mark]

[Total: 3 marks]

3b

(b)

List the known quantities:

- Total mass of the balloon and contents, $m = 1500 \text{ kg}$
- Gravitational field strength, $g = 9.81 \text{ N kg}^{-1}$

- Tension in each rope, $T = 400 \text{ N}$

(i) Calculate the weight of the balloon:

- Weight: $W = mg$ [1 mark]
- $W = 1500 \times 9.81 = 14\,715 \text{ N}$
- $W = 14\,700 \text{ N}$ [1 mark]

(ii) Calculate the resultant force acting on the balloon:

- Resultant force = upward forces – downward forces
- Resultant force = $2T$ OR $15\,500 - 14\,700$ [1 mark]
- Resultant force = 800 N [1 mark]

[Total: 4 marks]

For part (ii), the key points to realise are:

- When the ropes are tied to the ground, resultant force = 0
 - Resultant: $F_B - (W + 2T) = 0$
 - Buoyancy: $F_B = 14\,700 + (2 \times 400) = 15\,500 \text{ N}$
- When the ropes are untied, resultant force = $2T$, since the tension is no longer acting downwards
 - Resultant: $F_B - W = 2T$
 - Resultant: $15\,500 - 14\,700 = 800 \text{ N}$

3c

(c)

(i) Calculate the initial acceleration of the balloon:

List the known quantities:

- Mass of the balloon, $m = 1500 \text{ kg}$
- Initial resultant force, $F = 800 \text{ N}$

Substitute values into the acceleration equation:

- $F = ma$ [1 mark]

- $a = \frac{F}{m} = \frac{800}{1500}$

- $a = 0.5333 = 0.53 \text{ m s}^{-2}$ [1 mark]

(ii) The changes to the upwards acceleration during the first few seconds are:

- The speed (of the balloon) increases; [1 mark]
- The drag forces / downward forces (also) start to increase; [1 mark]

3c

(c)

(i) Calculate the initial acceleration of the balloon:

List the known quantities:

- Mass of the balloon, $m = 1500 \text{ kg}$
- Initial resultant force, $F = 800 \text{ N}$

Substitute values into the acceleration equation:

- $F = ma$ [1 mark]
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(ii) The changes to the upwards acceleration during the first few seconds are:

- The speed (of the balloon) increases; [1 mark]
- The drag forces / downward forces (also) start to increase; [1 mark]
- (Hence), acceleration decreases; [1 mark]

[Total: 5 marks]

We know that acceleration is the rate of change of velocity, so a simplistic approach would be to consider the motion as uniform acceleration.

However, in this situation, we have to consider the effects of drag. As velocity increases, drag forces act in the opposing direction to the motion of the balloon as it starts to move upwards.

Therefore, as drag forces increase, the **magnitude** of the acceleration decreases

3d

(d) When sand is released from the bags:

- The acceleration increases; [1 mark]

This is because:

Any **one** from:

- The upward / resultant force (on the balloon) increases; [1 mark]
- The downward force / weight (of the balloon) decreases; [1 mark]
- The mass (of the balloon) decreases; [1 mark]

[Total: 2 marks]

